

KUMAR AYUSH

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EDUCATION

New York University, New York, NY, USA

September 2023 - May 2025

Master of Science in Computer Science | GPA: 3.97/4.0

Relevant Coursework: Software Engineering, Design and Analysis of Algorithms, Machine Learning, Data Science, Big Data

Teaching Assistant for the Basic Algorithms and Java

SRM Institute of Science and Technology, Chennai, India

July 2019 - May 2023

Bachelor of Technology in Electronics and Communications Engineering | CGPA: 9.16/10

TECHNICAL SKILLS

Programming: C++, Python, C, Java, TypeScript, SQL, JavaScript | **Web Technologies:** React, Node.js, Django, RESTful APIs

Frameworks & Libraries: TensorFlow, Pandas, NumPy, Scikit-learn, Seaborn, Keras, PyTorch, OpenCV

LLM & Agents: LangChain, LangGraph, RAG, Vector DBs, Semantic Search, Multi-agent Systems, Chain-of-Thought, Embeddings

Data and Cloud: Amazon Web Services (AWS), Azure, Apache Spark, Hadoop, MySQL, MongoDB, PostgreSQL, NoSQL

DevOps and Tools: Git, Jenkins, Docker, Kubernetes, Agile, Jira, Linux/Unix, CI/CD Pipelines, Kafka

WORK EXPERIENCE

OkCupid (Contract), New York, USA

June 2025 - Present

Software Development Engineer

- Applied backend tech stack including **Node.js, PostgreSQL, Redis, and Elasticsearch** to build scalable microservices supporting millions of daily active users.
- Developed real-time pipelines with Kafka and batch processing with Apache Spark **to power accurate matchmaking and personalized recommendations**.
- Collaborated with product, AI, and data science teams to launch personalization **features that drove a 15% increase in daily matches and improved user retention by 10%**.

Ocean Shark, New Jersey, USA

May 2024 - December 2024

Software Development Engineer Intern

- Developed scalable **Node.js microservices on AWS**, reducing average response latency by **30% for core user transactions**.
- Designed a high-performance index and query system, integrating a cost-efficient storage solution that supported analytics workloads **at 20% lower storage cost**.
- Delivered **user-focused features in Agile sprints**, supporting a cloud-scale architecture **serving 500+ active users**.

EVE Healthcare, Gurugram, India

September 2021 - August 2023

Software Development Engineer

- Led **full-stack development** to build EVE Healthcare's web platform using **React, JavaScript** and **Django** integrating **RESTful APIs** in an **agile environment**.
- Integrated GitLab CI/CD pipelines enabling automated deployments across multiple environments, reducing deployment time.
- Achieved a 30% reduction in operational costs** by implementing **AWS cloud technologies** and optimizing backend systems with **Django**.

Samsung R&D Institute, Bangalore, India

February 2022 - November 2022

Samsung PRISM research intern

- Engineered a Python-based 5G MAC Control Decoder that improved network performance and **achieved 90% decoding accuracy**, aligning with industry benchmarks.
- Directed a cross-functional team of four, leveraging computer networking expertise to enhance decoding speed and efficiency.
- Collaborated with senior researchers to integrate the decoder into real-world testbeds, validating performance improvements and contributing to scalable 5G network solutions.

PROJECTS

Temporal Localization for Action Detection in Live Video Streams [[GitHub](#)] | *Apache Spark, Kafka, PyTorch, Deep Learning*

- Achieved 88.2% accuracy** on the **YouTube-8M Segment dataset** by designing a deep learning model tailored for identifying actions in specific video segments.
- Processed 1.71 TB of video data** using **TFRrecords** and **PyTorch DataLoader**, optimizing training speed via **Apache Spark** for distributed data processing.
- Created a **Kafka pipeline** to handle live streaming video input that performs real-time scene understanding.

AgentFlow - Dynamic AI-Human Work Allocation System [[GitHub](#)] | *LLM, RAG, FastAPI*

- Implemented an AI-driven workflow that automated **75% of task generation and reduced manual planning time by 60%**.
- Leveraged Meta-Llama-3 on RunPod's serverless GPUs for chat-based specification analysis, enabling scalable and cost-efficient AI inference.
- Implemented Retrieval-Augmented Generation (RAG) with FAISS embeddings for context retrieval and delivered a responsive React/Tailwind UI for real-time task visualization.

ACHIEVEMENTS

- Achieved 81st place in the **2022 ICPC Asia Taiwan** Online Programming Contest as part of Team Alpha (SRM Institute of Science and Technology).